Numerical Modeling of Aberration Effects in Mask-Based iSCAT Microscopy

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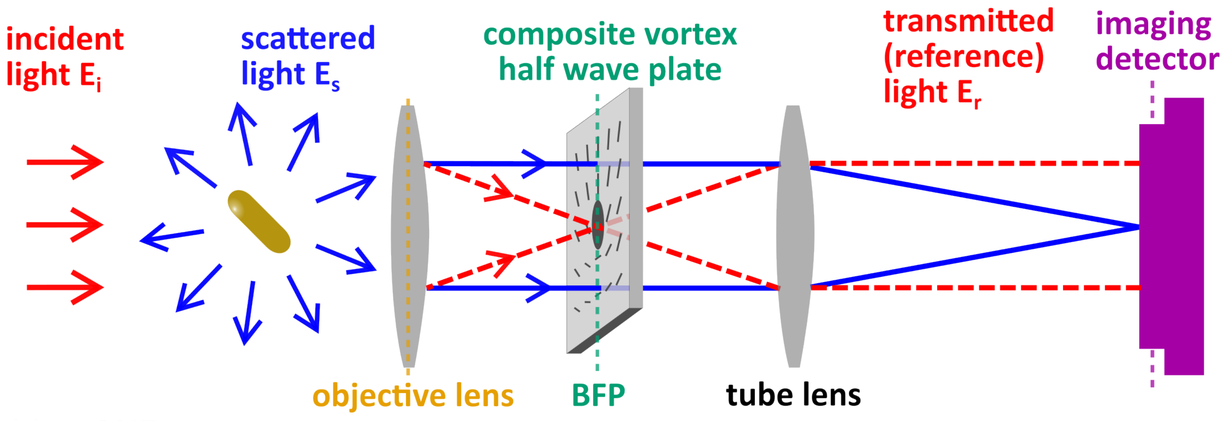
1 Introduction

Interferometric scattering (iSCAT) microscopy is an imaging technique that enables the detection and tracking of label-free nanoscale objects with high temporal resolution[1]. Recent methodological advances have extended iSCAT by introducing polarization-modulating masks in the back focal plane[2]. However, objective aberrations can introduce polarization distortions that interact with the mask and could lead to misinterpretation of experimental results.

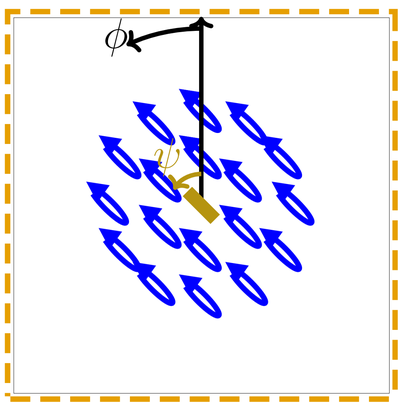
This work presents a systematic investigation and modeling of these effects, aiming to clarify their impact on images and improve the reliability of experimental interpretation.

2 Cylindrically polarized iSCAT (cypiSCAT)

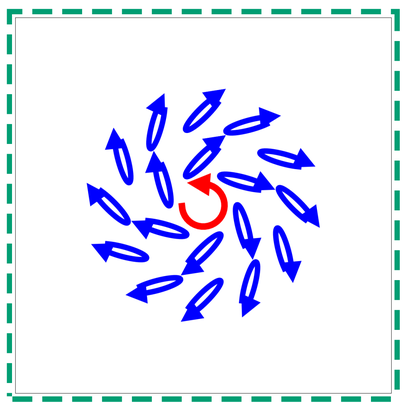
Cylindrically polarized iSCAT (cypiSCAT) enables the measurement of nanoscale rotations[2]. This is achieved by introducing a composite vortex half-wave plate that transforms the scattered light in the back focal plane into a cylindrically polarized field, producing a dipole-like interference pattern in the image plane that rotates together with the anisotropic scatterer.



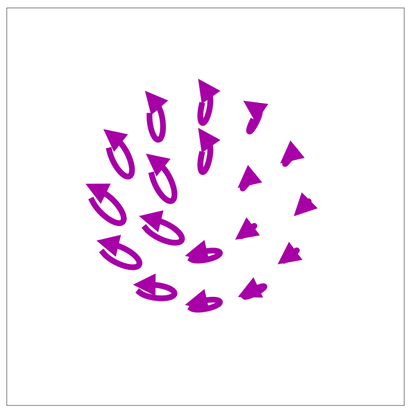
(a) cypiSCAT setup



(b) Scattered field (blue) and anisotropic scatterer orientation (gold)



(c) Back Focal Plane (BFP) field: Scattered (blue) and Reference (red)



(d) Total field (purple)

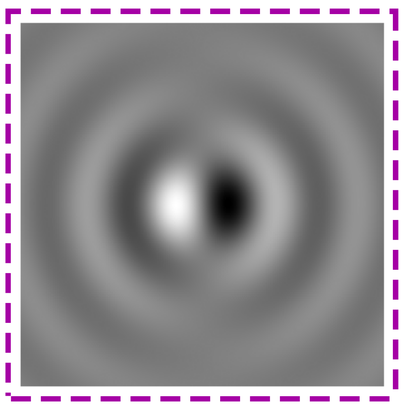
$$\mathbf{M}_m^{\mathrm{CylPol}}(\phi) = \begin{pmatrix} \cos(m\phi) & \sin(m\phi) \\ \sin(m\phi) & -\cos(m\phi) \end{pmatrix} \qquad \vec{E}_{\mathrm{inc}}^{\mathrm{LHC}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\mathbf{R}(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix} \qquad \boldsymbol{\alpha}(\psi) = \mathbf{R}(\psi)\begin{pmatrix} \alpha_x & 0 \\ 0 & \alpha_y \end{pmatrix}\mathbf{R}(-\psi)$$

$$I \sim \left|\left|\left(\mathbf{M}_{m=\pm1}^{\mathrm{CylPol}}(\phi)\,\boldsymbol{\alpha}(\psi) + \mathbf{I}\right)\vec{E}_{\mathrm{inc}}^{\mathrm{LHC}}\right|\right|^2 \sim (\alpha_x - \alpha_y)\cos(2\psi - \phi)$$

$\Rightarrow$ Dipole pattern rotates with the scatterer[2]

where $\mathbf{M}_m^{\mathrm{CylPol}}(\phi)$ is the cylindrical mask transform, $\vec{E}_{\mathrm{inc}}^{\mathrm{LHC}}$ is the incident field, $\mathbf{R}(\psi)$ is a rotation matrix, α_x and α_y define scatterer's polarizability, ϕ is the azimuthal angle, ψ is the scatterer's orientation and $\mathbf{I}$ is the identity matrix.

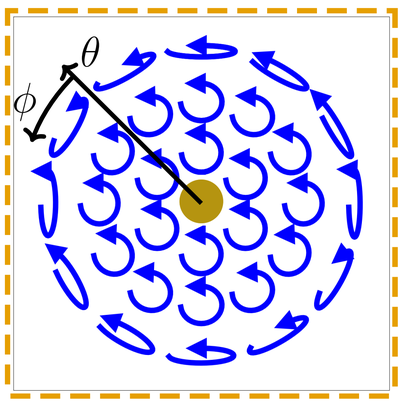


(e) Intensity in the image plane, showing a dipole pattern

3 Isotropic scatterers in cypiSCAT and the effect of objective aberrations

Let us now examine the dominant effects in image formation for an isotropic scatterer.

Here, depolarization effects in optical components, mainly the objective[3], play a critical role.



(a) Aberrated scattered field (blue) of an isotropic particle (gold)

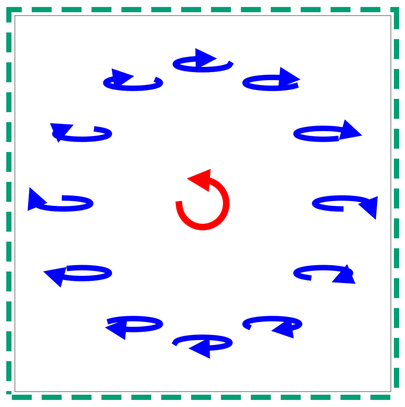
Polarization is distorted according to Fresnel coefficients and coordinate change (described by $\mathbf{P}$) on the objective:

$$\mathbf{P}(\theta,\phi) = \mathbf{R}(\phi)\begin{pmatrix} t_{\mathrm{p}}(\theta)\cos\theta' & 0 \\ 0 & t_{\mathrm{s}}(\theta) \end{pmatrix}\mathbf{R}(-\phi)$$

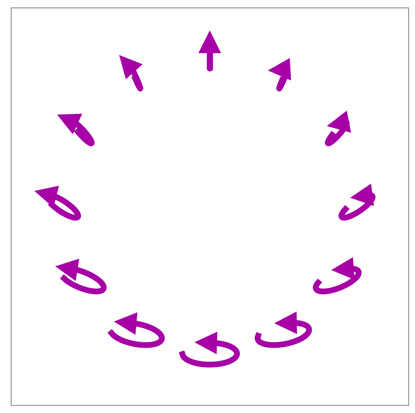
$$\cos\theta' = \sqrt{1 - \left(\frac{n_{\mathrm{oil}}\sin\theta}{n_{\mathrm{air/water}}}\right)^2},$$

where $\mathbf{R}(\phi)$ is a rotation matrix, ϕ is the azimuthal angle, θ is the ray direction angle, $t_{\mathrm{p}}(\theta)$, $t_{\mathrm{s}}(\theta)$ are Fresnel transmission coefficients.

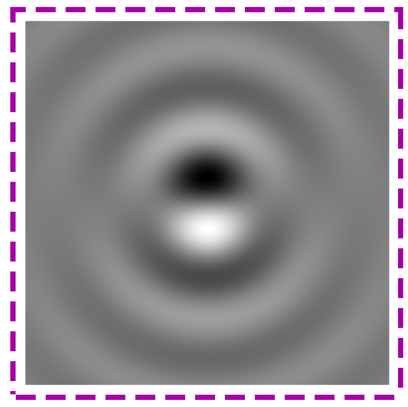
$$m = 1 \;\rightarrow\; I \sim \left|\left|\left(\mathbf{M}_{m=1}^{\mathrm{CylPol}}(\phi)\,\mathbf{P}(\theta,\phi) + \mathbf{I}\right)\vec{E}_{\mathrm{inc}}^{\mathrm{LHC}}\right|\right|^2 \sim \cos(\phi) \;\Rightarrow\; \text{Dipole pattern}$$



(a) Back Focal Plane (BFP) field: Scattered (blue) and Reference (red)

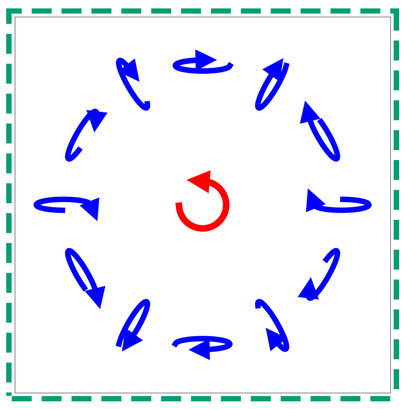


(b) Total field (purple)

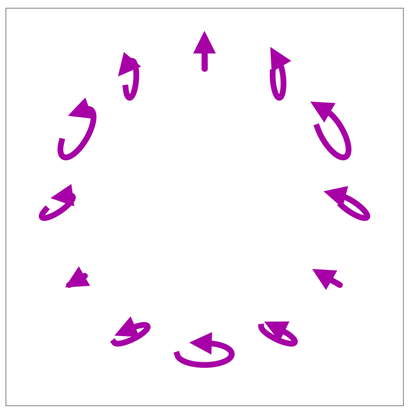


(c) Intensity in the image plane, showing a dipole pattern

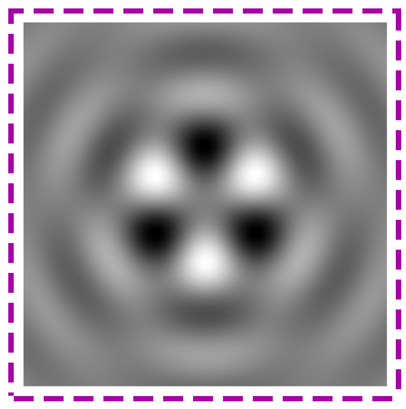
$$m = -1 \;\rightarrow\; I \sim \left|\left|\left(\mathbf{M}_{m=-1}^{\mathrm{CylPol}}(\phi)\,\mathbf{P}(\theta,\phi) + \mathbf{I}\right)\vec{E}_{\mathrm{inc}}^{\mathrm{LHC}}\right|\right|^2 \sim \cos(3\phi) \;\Rightarrow\; \text{Hexapole pattern}$$



(d) Back Focal Plane (BFP) field: Scattered (blue) and Reference (red)



(e) Total field (purple)



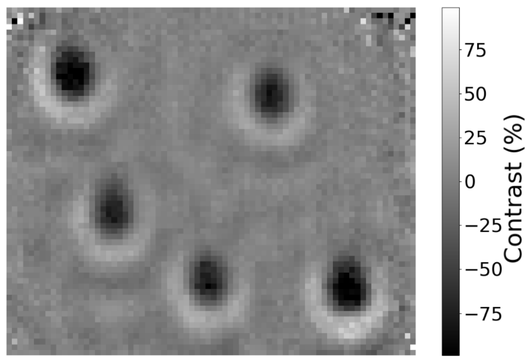
(f) Intensity in the image plane, showing a rotational symmetry

Theoretical contrasts of 20nm gold nanoparticles (GNP) or gold nanorods (GNR) of equivalent volume, in reflection mode, NA=1.3, on water:

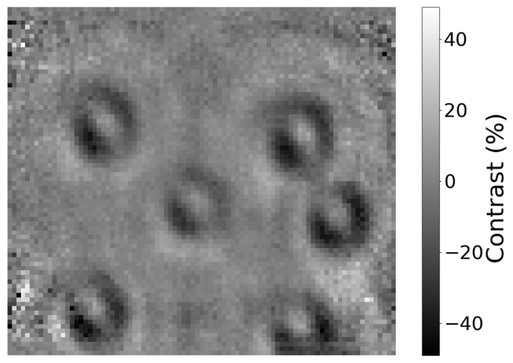
Theoretical contrast	$m = 1$	$m = -1$
GNP (fixed pattern)	1.8%	1.2%
GNR (rotating dipolar PSF)	3.9%	3.9%

4 Experimental validations

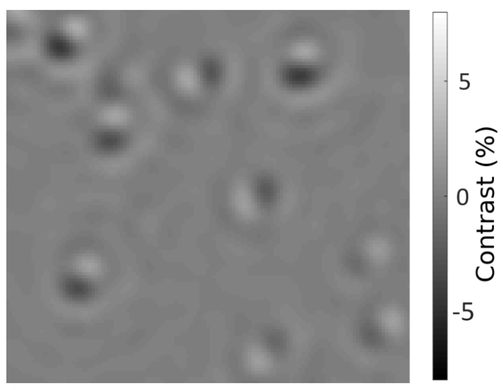
Measured images of spherical 60nm gold nano-particles (GNPs, (a,b,d,e)) or equivalent gold nano-rods (GNRs, (c)) on water, NA=1.3, in reflection mode (a,b) or transmission mode (c,d,e).



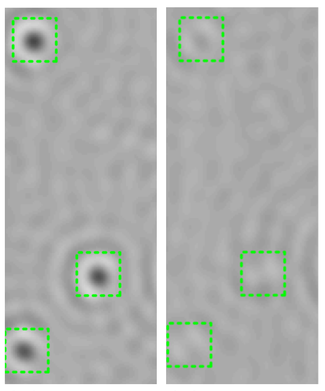
(a) No mask, GNP, reflection mode; standard iSCAT PSF



(b) $m = 1$, GNP, reflection mode; fixed dipolar PSF



(c) $m = 1$, GNR, transmission mode; dipolar PSF with varying orientations [2]



(d) No mask, GNP [2]

(e) m=1, GNP, transmission [2]

5 Conclusion

- Polarization aberrations in the objective, combined with a cylindrical polarization mask, generate fixed background patterns in the images of scatterers.
- Orientation retrieval for weakly anisotropic scatterers may require subtraction of these patterns.
- A vortex half-wave plate with $m = -1$ is expected to reduce the contrast of fixed patterns, enabling orientation retrieval of less anisotropic particles.
- The presented numerical model provides a foundation for the design and optimization of back focal plane masks.

6 References

(1) Ginsberg, N. S., Hsieh, C.-L., Kukura, P., Piliarik, M., & Sandoghdar, V. (2025). Interferometric scattering microscopy. Nature Reviews Methods Primers, 5(1). https://doi.org/10.1038/s43586-025-00391-1

(2) Vala, M., Kopal, I., Takiguchi, L., Shaidiuk, Y., Lužný, V., Bujak, Ł., Kosuri, P., & Piliarik, M. (2026). Measuring Nanoscale Torques with Cylindrical-Polarization-based Interferometric Scattering Microscopy. arXiv. https://doi.org/10.48550/arxiv.2601.08373

(3) Lin, S., Jiao, N., Shaidiuk, Y., Feng, D., Luo, J., Yu, Y., Bujak, L., Tang, J., Piliarik, M., & Chen, X. (2025). Photothermal fourier-plane phase synchronization for interferometric scattering microscopy. arXiv. https://doi.org/10.48550/arxiv.2510.15468

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